

Nathan Potraz

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SUMMARY

Results-driven Software Engineer with 4+ years of experience architecting full-stack solutions. Expertise in Java, Go, and Python backend development, and React/TypeScript frontend. Seeking a Software Engineering role to build innovative solutions.

EXPERIENCE

Hardware Specialist & Small Business Owner

Self-Employed

June 2022 - October 2024, Oceanside, CA

- Engineered and fabricated over **100** custom-tuned competitive gaming controllers, earning an average **5/5 rating across 76 customer reviews**.
- Developed and implemented **intricate modification protocols** to enable advanced features (e.g stick-profile tuning and button remapping) on custom hardware, **demonstrating complex system integration**.
- Optimized device hardware via precision shell modifications to **achieve specific input values out of a total of 352 values** - critical for high-level competitive play.

PROJECT

Devlog

devlog.ntpotraz.dev/ · October 2025 - Present

- Architected a **high-speed backend** (Golang/SQLite/Turso) to enable instantaneous create, edit, and retrieval of log entries. **Sub 1 second** database reads and writes.
- **Implemented CI/CD using GitHub Actions and Docker** to automate building and deploying to Google Cloud Run when a new feature gets merged to main.
- Integrated Clerk for user authorization, supporting SSO (GitHub/Google) and email/password login, **scalable from 1,000s of users to 1,000,000s**.

Custom DNS Resolver Simulation

github.com/Saltair1/NetworkPythonProject/ · October 2023 - November 2023

- Developed a full DNS Client-Server ecosystem in Python, capable of resolving standard queries (A, AAAA, CNAME, NS) through a **recursive lookup chain**.
- Engineered custom packet construction logic using **bitwise operations** to manually encode and decode DNS headers, flags, and transaction IDs without relying on high-level DNS libraries.
- Implemented a recursive forwarding engine that checks local caches before routing specific domain requests to their respective authoritative servers.
- Built a Time-To-Live (TTL) management system that automatically tracks record age and purges expired entries from the client and server caches, utilizing **Pandas for real-time visualization** of Resource Record tables.

Hack Compiler

github.com/ntpotraz/VMTranslator/ · github.com/ntpotraz/Assembler/ · March 2020 - April 2020

- Engineered a complete **compilation chain in Java** to translate high-level Virtual Machine code into executable 16-bit binary, utilizing modular **Parser and CodeWriter** logic for efficient instruction mapping and file I/O.
- Developed a **two-pass Assembler** that leveraged a **custom Symbol Table data structure** to successfully resolve user-defined labels and pre-defined symbols before generating the final machine code.
- Built a VM Translator to map stack arithmetic and memory access commands into **optimized Hack Assembly instructions**, validating the entire system against a rigorous test suite to ensure bit-perfect output.

EDUCATION

B.S in Computer Science

California State San Marcos · San Marcos, CA · 3.7 GPA

Relevant Coursework: OOP, Data Structures & Algorithms, Networking, Cloud, Web Programming, Databases, AGILE/SCRUM.

SKILLS

Languages: Java, Go, Python, JavaScript/TypeScript, React, SQL

Tools: Git, GitHub, Linux, Agile/SCRUM, AWS, GCP, Docker, CI/CD
